

PROJECT PROPOSAL

IT138IU – Data Science and Data Visualization

Resource and Energy Sustainability: Vietnam vs. 7 Southeast Asian Countries

No.	Role	Name	Student ID
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Repository: <https://github.com/LeXuanQue/sea-resource-sustainability-viz>

1. Background and motivation

The World Bank's Sovereign ESG framework assesses environmental sustainability through natural-resource management and sustainable energy use [1]. For Viet Nam this is national policy: the National Climate Change Strategy targets net-zero emissions and 43% forest cover by 2050 [2]. (The World Bank records 47.2% forest area in 2022 [1]; the two use different forest definitions and are not directly comparable.)

According to Ember, Viet Nam's electricity demand grew by an average of 8.2% per year between 2013 and 2024, with coal-based generation expanding to meet it while wind and solar also grew [3]; coal still supplied 45% of ASEAN electricity in 2024 [3]. Viet Nam's position only becomes clear against its neighbours, and with 12 indicators, 8 countries and three decades of data, an interactive interface lets users choose the indicator, country and year rather than read one composite score.

2. Objectives

We will build an interactive web visualization comparing Viet Nam with seven Southeast Asian countries on 12 natural-resource and energy indicators, answering:

- RQ1: How have Viet Nam's 12 natural-resource and energy indicators changed over time?
- RQ2: How does Viet Nam compare with Indonesia, Thailand, Malaysia, the Philippines, Cambodia, Laos and Myanmar across the 12 indicators over time?
- RQ3: How does the gap between Viet Nam and the selected-country regional median change over time for the 12 indicators?

3. Data

Data source: the World Bank Sovereign ESG dataset [1], downloaded once as a CSV (239 economies, 71 indicators, 1960–2023), so the data is fixed and reproducible. We keep 8

countries and 12 indicators from two Environment themes; indicator codes are listed in the repository README.

Natural resources: forest area (% of land area), tree cover loss (hectares), protected areas (% of territory), resource depletion (% of GNI), net forest depletion (% of GNI), freshwater withdrawals (% of internal resources).

Energy: energy intensity (MJ/\$2017 PPP GDP), renewable energy (% of final energy), renewable electricity (% of electricity), energy use per person (kg oil equivalent), fossil fuel use (% of total energy), electricity from coal (% of electricity).

Coverage: Most series run from 1990 to 2021 or 2022; tree cover loss covers 2002–2021, protected areas 2013–2023 and energy intensity 2000–2022. Cambodia starts in 1995 on most series and Laos in 2000 on energy use, fossil fuel and coal; freshwater withdrawals start in 2005–2007 for four countries; Laos reports renewable electricity for only 10 scattered years.

4. Data processing

A Python script converts the CSV into three static JSON files (values, latest snapshot, coverage); the browser only filters.

- Reshape: wide to long form, one row per country, indicator, year and value.
- Missing data: stored as null, never 0; charts never interpolate across gaps. Repeated estimates (Viet Nam's freshwater withdrawals stay at 22.78% from 2005 to 2021) are drawn lighter.
- Year shown: the latest year with at least 6 of 8 countries reporting, printed beside every value.
- Regional reference: the median, since one country can drag the mean far (freshwater: mean 11.0%, median 7.4%).
- Direction and rank: each indicator is tagged higher- or lower-is-better, so rank 1 is always best; energy use per person has no agreed direction and no rank.
- Verdict: better or worse against the median, level within 3%; change since 2010 is judged separately.

Limitations: tree cover loss is in hectares, favouring smaller countries; protected areas has no 2010 baseline; values are national averages.

5. Visualization design

All three designs answer the same question with the same three levels (region, one country, one indicator over time) and differ in how the reader moves between them. The six sketches are in the appendix.

5.1 Design A: story first, then explore

Following the "martini glass" structure of narrative visualization [4], a hero gives the question, a one-sentence answer and three headline numbers (forest area +18.4 points since 1990, renewable energy share −51.7 points, 2.9% of territory protected), followed by three story blocks (Figure 1). An explore section then links a ranked bar list and map, a dot plot of all 12 indicators for one country, and a trend line against the regional median; clicking a

country drills down, a back control returns (Figure 2). Strength: the main message needs no interaction. Weakness: a long page whose story text depends on the data.

5.2 Design B: coordinated dashboard

One screen with shared controls, a KPI strip, a ranked bar list, six small multiples and a trend chart; clicking a country updates every panel (Figures 3–4). Strength: every level visible at once. Weakness: the message needs interaction, and six panels are dense.

5.3 Design C: head-to-head comparison

Three numbered full-page levels: a map with a ranking, a country compared with one partner country, and a multi-line trend with a coverage table (Figures 5–6). Strength: the clearest path between levels and thorough missing-data labels. Weakness: it benchmarks against the regional average, conflicting with our median rule, and is the largest build.

5.4 Chosen direction

Criterion	Design A	Design B	Design C
Main message readable without interaction	Yes	No	Partly
Three levels with a path between them	Drill-down and back	All on one screen	Most explicit
Shows missing data explicitly	Yes	Yes	Yes
Build effort in 10 weeks, no prior frontend	Medium	Medium to high	High

We will implement Design A: it is the only design whose main finding is readable without interaction, it still covers every research question through its explore section, and it is the lowest-risk build for a team new to frontend work.

6. Must-have features

- M1 Region level: ranked bar list of the 8 countries, regional median marked.
- M2 Country level: all 12 indicators for one country, with rank and verdict.
- M3 Indicator level: trend line from 1990 against the dashed regional median.
- M4 Shared controls: indicator, country and year, updating every panel.
- M5 Drill-down: click a country to open its view; a back control returns.
- M6 Tooltips: country, year, value and unit on every mark.
- M7 Missing data: shown as "no data", line breaks or hatching, never zero.
- M8 Provenance: units on every chart and the data source on the page.
- M9 Delivery: static site on GitHub Pages.
- M10 Story section: hero and three story blocks, readable without interaction.

RQ1 is answered by M2, M3 and M10; RQ2 by M1, M4 and M5; RQ3 by M2, M3 and M10.

7. Optional features

Orientation map, collapsible details table, animated transitions, shareable links, and a Vietnamese interface, if time allows after week 7.

8. Project schedule

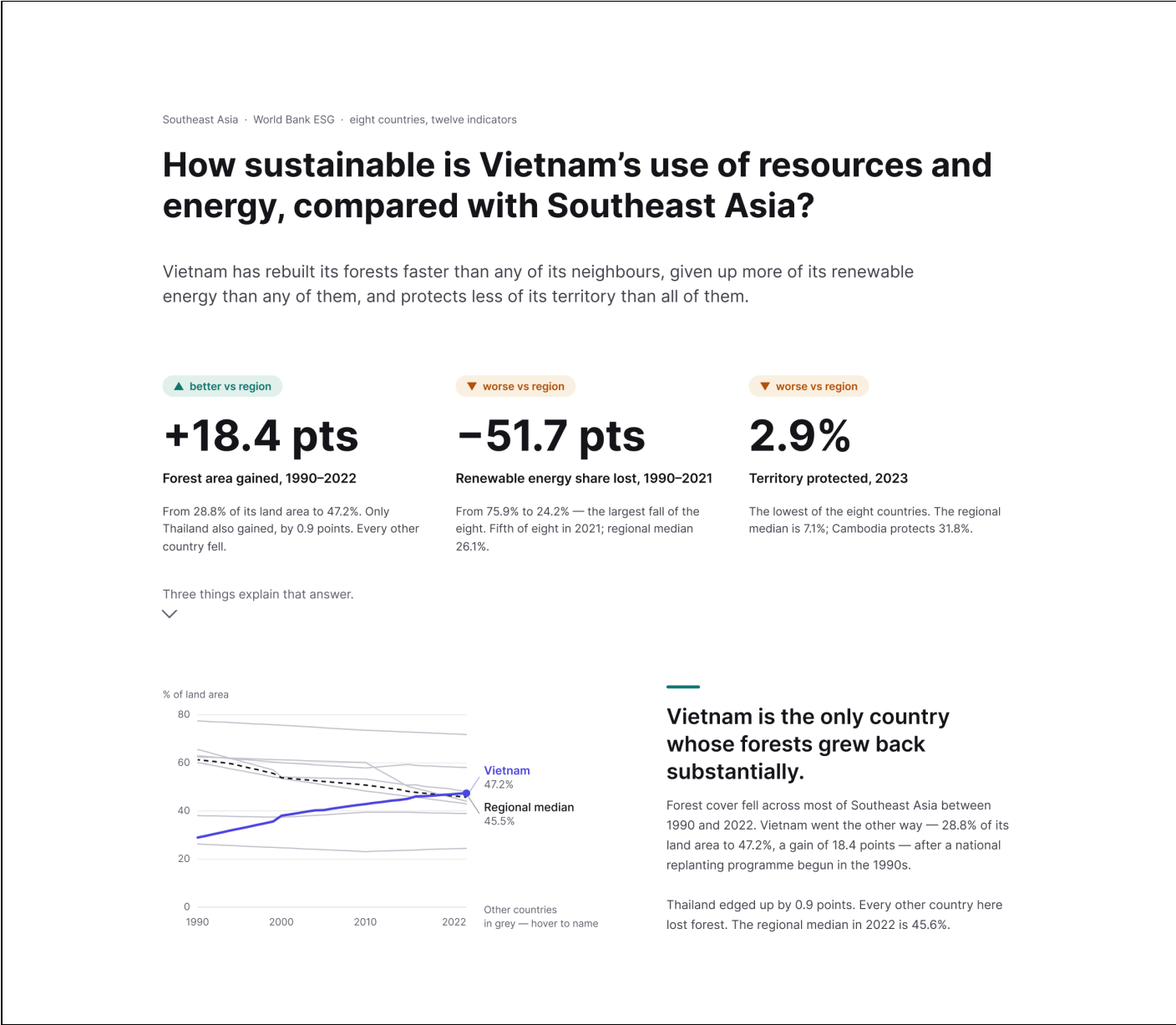
Ten weeks from 5 October 2026 (full plan: docs/project-schedule.md).

Week	Milestone
1–2	Kickoff, proposal, design choice, JSON schema, page skeleton
3–4	JSON v1; region, country and indicator levels; shared controls; drill-down
5–6	Midterm integration and bug fixes; working prototype due (week 6)
7–8	Feedback, improvements, optional features, report sections
9–10	Feature freeze, testing, report; final submission (week 10)

Weeks 3–4 are led by Trần Nguyễn Lê Quân (data), Lê Xuân Quế and Đặng Hoàng Quân (frontend); all other weeks are shared. Weeks 1–2 include time to learn D3.js, as no member has built a frontend before.

References

- [1] World Bank. Sovereign ESG Data Framework.
<https://esgdata.worldbank.org/data/framework>, accessed 2 October 2026.
- [2] Chính phủ Việt Nam. 2022. Chiến lược quốc gia về biến đổi khí hậu đến năm 2050 (National Climate Change Strategy to 2050).
<https://xaydungchinhhsach.chinhphu.vn/phe-duyet-chien-luoc-quoc-gia-ve-bien-doi-khi-hau-dien-nam-2050-119220727070101019.htm>, accessed 2 October 2026.
- [3] Ember. From emission-intensive to investment hotspots: championing renewables in 3 ASEAN economies.
<https://ember-energy.org/latest-insights/from-emission-intensive-to-investment-hotspots-championing-renewables-in-3-asean-economies/country-snapshots-and-opportunities/>, accessed 2 October 2026.
- [4] Segel, E. and Heer, J. 2010. Narrative Visualization: Telling Stories with Data. IEEE Transactions on Visualization and Computer Graphics 16(6).



Click a country (drill-down)

Clicking Laos switches every panel to Laos. Vietnam stays on screen as a reference. Renewable electricity also shows the gap rule: Laos reports ten scattered years, and the isolated 2012 reading is a single dot.

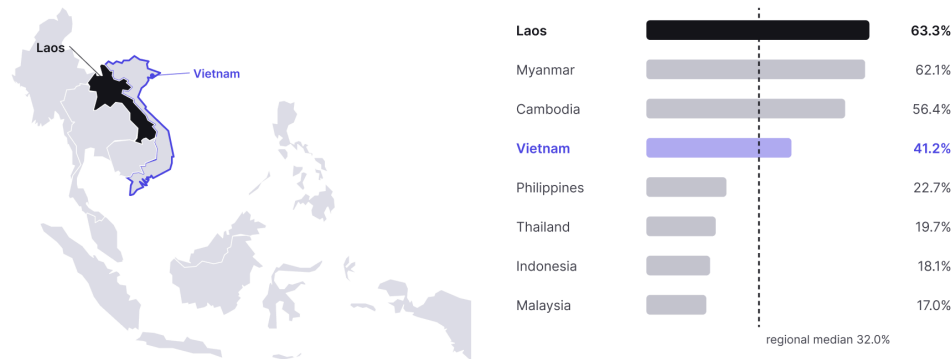
Now explore the rest yourself.

One indicator, one country, one year — shared by all three panels. The year starts at the latest the indicator reports.

Indicator **Renewable electricity** Country **Laos** Year **2021** 1990 2021 [← Back to Vietnam](#)

Laos gets the largest share of its electricity from renewables.

% of electricity, 2021 · best at top · dashed line is the median of the 8 countries reporting



The map is here for orientation — the values are in the list.

↓ click a country

Drag the year to 1995

Every panel follows the year. In 1995 four countries have no freshwater reading, and three whole indicators do not start until later.

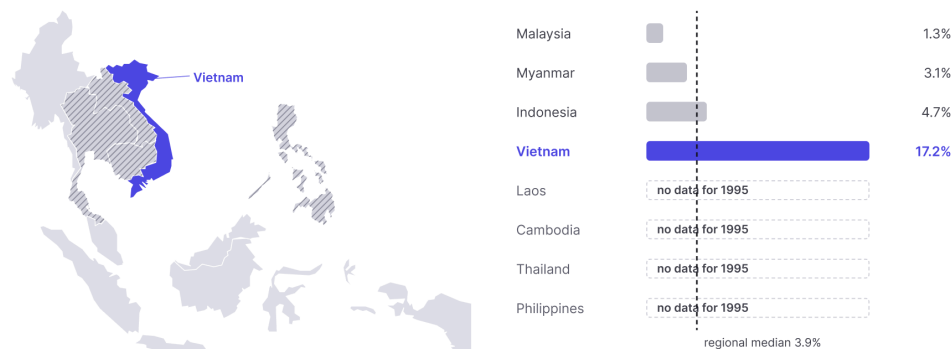
Now explore the rest yourself.

One indicator, one country, one year — shared by all three panels. The year starts at the latest the indicator reports.

Indicator **Freshwater withdrawals** Country **Vietnam** Year **1995** 1990 2021

Only four of the eight countries reported freshwater withdrawals in 1995.

% of internal resources, 1995 · best at top · dashed line is the median of the 4 countries reporting



The map is here for orientation — the values are in the list.

Figure 2. Design A: clicking a country re-ranks and highlights it; years with no data are labelled, never drawn as zero.

SEA Sustainability

Vietnam vs 7 Southeast Asian countries · World Bank Sovereign ESG

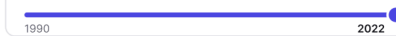
INDICATOR

Forest area (% of land area)

COUNTRY

Vietnam

YEAR



Natural resources

Energy

vs regional median · latest year per indicator

Forest area

47.2

% of land area

▲ better

median 45.6 · 2022

Tree cover loss

193k

hectares

▲ better

median 235k · 2021

Protected areas

2.9

% of territory

▼ worse

median 7.05 · 2023

Resource depletion

0.51

% of GNI

▲ better

median 2.31 · 2021

Net forest depl.

0

% of GNI

= level

median 0 · 2021

Freshwater use

22.8

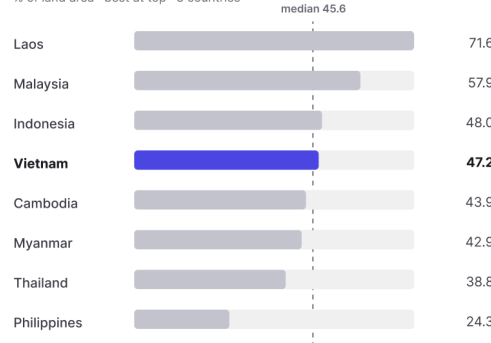
% of internal resources

▼ worse

median 7.44 · 2021

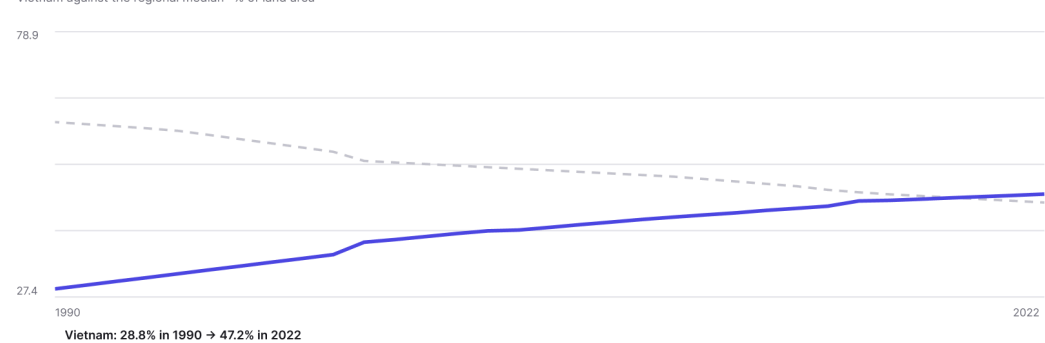
Forest area, 2022

% of land area · best at top · 8 countries



Forest area over time, 1990–2022

Vietnam against the regional median · % of land area

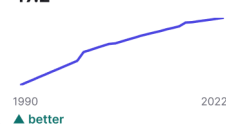


ALL SIX NATURAL-RESOURCE INDICATORS FOR VIETNAM

Forest area

47.2

% of land area



Tree cover loss

193k

hectares



Protected areas

2.9

% of territory



Resource depletion

0.51

% of GNI



Net forest depl.

0

% of GNI



Freshwater use

22.8

% of internal resources



Regional reference is the MEDIAN of the 8 countries, never the mean. Missing values are shown as "no data", never 0. Ranks are direction-adjusted, so rank 1 is always the best outcome.

Source: World Bank Sovereign ESG · esgdata.worldbank.org

Figure 3. Design B: coordinated dashboard, every view on one screen.

SEA Sustainability

Vietnam vs 7 Southeast Asian countries · World Bank Sovereign ESG

INDICATOR

Forest area (% of land area)

COUNTRY

Laos

YEAR

1990 2022

Natural resources

Energy

vs regional median · latest year per indicator

Laos selected x

Forest area

71.6

% of land area

▲ better

median 45.6 · 2022

Tree cover loss

318k

hectares

▼ worse

median 235k · 2021

Protected areas

19.0

% of territory

▲ better

median 7.05 · 2023

Resource depletion

3.44

% of GNI

▼ worse

median 2.31 · 2021

Net forest depl.

1.59

% of GNI

▼ worse

median 0 · 2021

Freshwater use

3.86

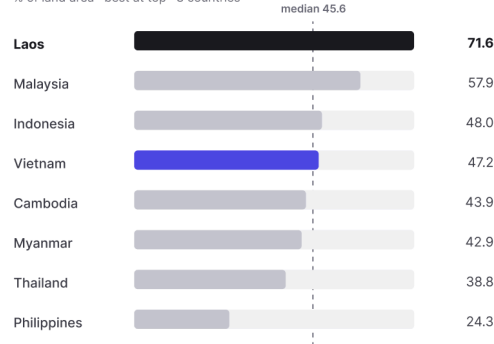
% of internal resources

▲ better

median 7.44 · 2021

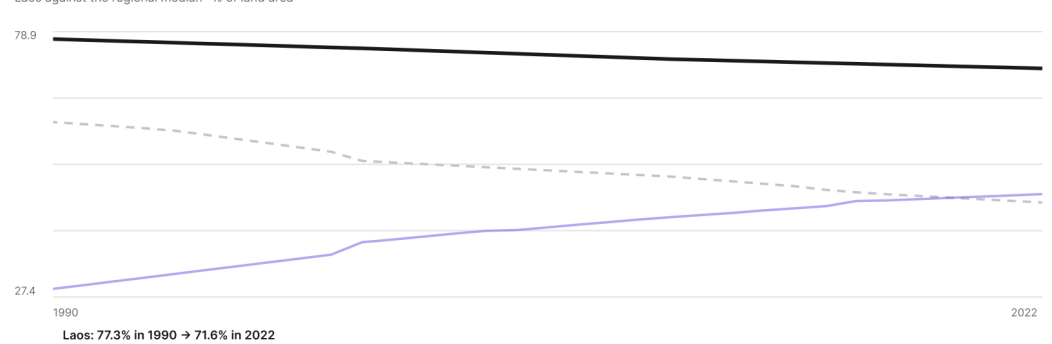
Forest area, 2022

% of land area · best at top · 8 countries



Forest area over time, 1990–2022

Laos against the regional median · % of land area



ALL SIX NATURAL-RESOURCE INDICATORS FOR LAOS

Forest area

71.6

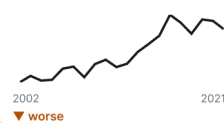
% of land area



Tree cover loss

318k

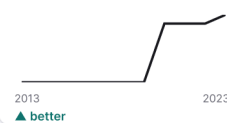
hectares



Protected areas

19.0

% of territory



Resource depletion

3.44

% of GNI



Net forest depl.

1.59

% of GNI



Freshwater use

3.86

% of internal resources



Regional reference is the MEDIAN of the 8 countries, never the mean. Missing values are shown as "no data", never 0. Ranks are direction-adjusted, so rank 1 is always the best outcome.

Source: World Bank Sovereign ESG · esgdata.worldbank.org

Figure 4. Design B: one click updates the KPI strip, small multiples and trend together.

VIEW

Map + ranking



% of land area · World Bank ESG

8 countries · annual observations
1990–2022, coverage varies

Gray hatching = no data (never zero)

REGIONAL DISTRIBUTION

Country ranking

Selected year · higher values shown first

2022



REGIONAL AVERAGE

46.8%

Unweighted mean

REGIONAL MEDIAN

46.5%

Midpoint of 8 peers

VIETNAM VS REGIONAL AVERAGE

Near the regional center in 2022

+0.5 pp

Figure 5. Design C, level 1: controls and a ranking of all eight countries, with average and median cards.



click an indicator card

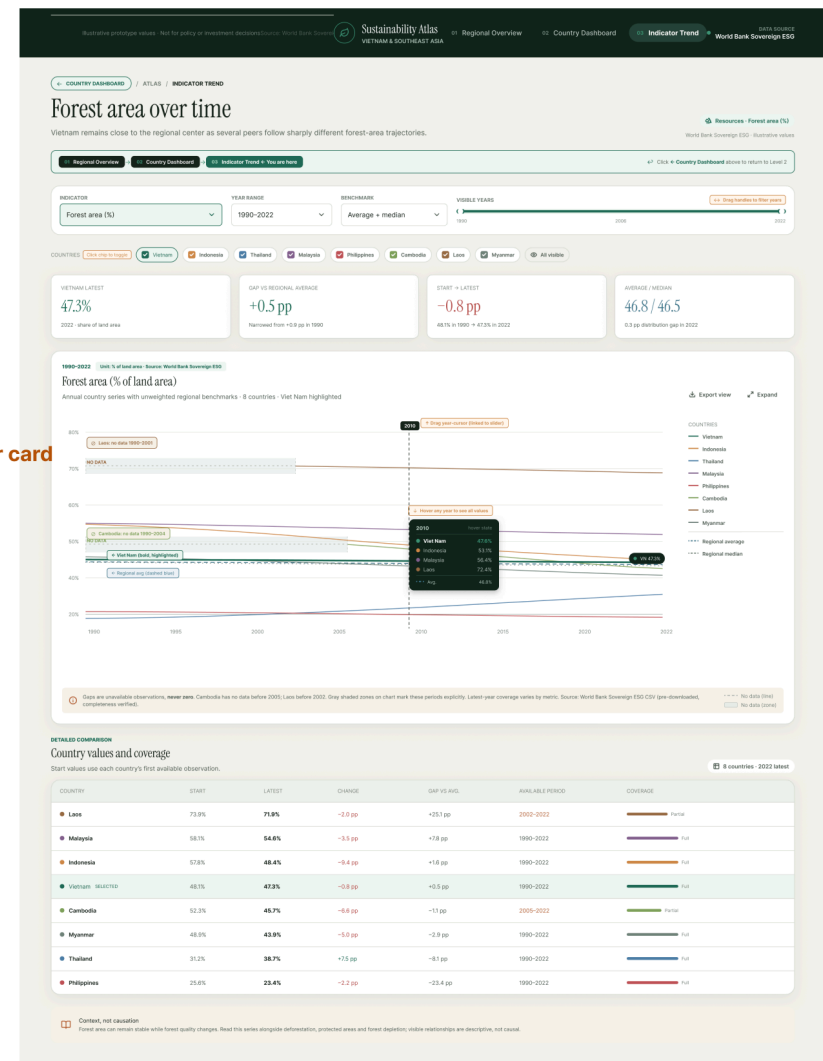


Figure 6. Design C: clicking an indicator card opens its full trend; gaps are named, never drawn as zero.